Systems of Equations Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Substituting an expression

ORIGINAL PROBLEM

$y = 2x - 1$ and $3x + 2y = 12$

INCORRECT WORKED SOLUTION

$3x + 2y = 12$

$3x + 2 \cdot 2x - 1 = 12$

$7x - 1 = 12$

$7x = 13$

$x = \frac{13}{7}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Finishing the system

ORIGINAL PROBLEM

$x + y = 7$ and $x - y = 1$

INCORRECT WORKED SOLUTION

Add the equations: $2x = 8$

$x = 4$

Solution: $x = 4$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Systems of Equations Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

3. Subtracting equations

ORIGINAL PROBLEM

$5x + 3y = 21$ and $2x + 3y = 12$

INCORRECT WORKED SOLUTION

Subtract the equations:

$3x = 21 + 12$

$3x = 33$

$x = 11$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Multiplying the whole equation

ORIGINAL PROBLEM

$x + 2y = 8$ and $3x - y = 3$

INCORRECT WORKED SOLUTION

Multiply the second equation by 2:

$6x - 2y = 3$

Add to the first: $7x = 11$

$x = \frac{11}{7}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Systems of Equations Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

5. Parallel lines

ORIGINAL PROBLEM

$y = 2x + 3$ and $y = 2x - 1$

INCORRECT WORKED SOLUTION

$2x + 3 = 2x - 1$

$3 = -1$

The x terms cancelled, so $x = 0$.

$y = 2(0) + 3 = 3$. Solution: (0, 3)

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Reading the result

ORIGINAL PROBLEM

$2x + 4y = 8$ and $x + 2y = 4$

INCORRECT WORKED SOLUTION

Multiply the second equation by -2:

$-2x - 4y = -8$

Add: $0 = 0$

No solution

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Systems of Equations Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

7. Checking a solution

ORIGINAL PROBLEM

Is (2, 1) a solution of x + y = 3 and 2x − y = 4?

INCORRECT WORKED SOLUTION

Check x + y = 3: 2 + 1 = 3 ✓

The point works, so (2, 1) is a solution.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. Setting up equations

ORIGINAL PROBLEM

Adult tickets cost $8 and student tickets cost $5. 40 tickets were sold for $275. Write a system.

INCORRECT WORKED SOLUTION

a = adult tickets, s = student tickets

a + s = 275

8a + 5s = 40

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Systems of Equations Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Ordered pair

ORIGINAL PROBLEM

$x + y = 10$ and $x - y = 4$

INCORRECT WORKED SOLUTION

Add: $2x = 14$, so $x = 7$

$7 + y = 10$, so $y = 3$

Solution: (3, 7)

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Combining like terms

ORIGINAL PROBLEM

$x = 3 - y$ and $2x + 3y = 8$

INCORRECT WORKED SOLUTION

$2(3 - y) + 3y = 8$

$6 - 2y + 3y = 8$

$6 + 5y = 8$

$5y = 2$

$y = \frac{2}{5}$

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points